2025 Erich J. Spengler Midwest Regional Collegiate Cyber Defense Competition Team Packet

March 14-15, 2025

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Midwest Regional CCDC Mission and Objectives

The 2025 Erich J. Spengler Midwest Regional Collegiate Cyber Defense Competition (CCDC) provides a competitive opportunity for collegiate teams who have proven themselves in 2025 Midwest State CCDC Qualification or Wildcard events. The Midwest Regional Collegiate Cyber Defense Competition is designed to provide a controlled competitive environment that will permit each participating institution to assess their students' depth of understanding and operational competency in managing the challenges inherent in protecting an enterprise network infrastructure and business information systems. The winner of the 2025 Erich J. Spengler Midwest Regional CCDC is eligible to move on to the 2025 National CCDC on April 25-27, 2025. The 2025 National CCDC has been announced as an all virtual event. The second place team of the regional is eligible to compete in the National CCDC Wildcard competition on April 9, 2025, an all virtual event. The winner of the National CCDC Wildcard is eligible to move on to the 2025 National CCDC

Overview

Midwest Collegiate Cyber Defence Competitions are managed by the MWCCDC Consortium, an LLC independent from Moraine Valley Community College. The competition is designed to test each student team's ability to secure a networked computer system while maintaining standard business functionality.

The 2025 Erich J. Spengler Midwest Regional CCDC will be hosted at Purdue University Northwest in Hammond, IN. The Hammond campus is at,

2200 169th Street, Hammond, IN 46323

The teams are expected to manage the computer network, keep it operational, and prevent unauthorized access. Each team will be expected to maintain and provide public services per company policy and mission. Each team will start the competition with a set of identically configured systems.

This is not just a technical competition, but also one built upon the foundation of business operations, policy, and procedures. A technical success that adversely impacts the business operation will result in a lower score as will a business success which results in security weaknesses.

Student teams will be scored on the basis of their ability to detect and respond to outside threats, including cyber attacks while maintaining availability of existing network services such as mail and web servers, respond to business requests such as the addition or removal of additional services, and balance security against varying business needs.

There are ten teams competing in the 2025 Erich J. Spengler Regional CCDC comprising eight winners of respective state qualification CCDC, plus two wildcard teams.

Participating teams in the 2025 Erich J. Spengler Midwest CCDC are, in alphabetic order,

Baker College
Baldwin Wallace University
Cedarville University
DePaul University
Eastern Kentucky University
Indiana Institute of Technology
Missouri State University
Southeast Missouri State University
St. Cloud State University
University of Wisconsin-Stout

Business Scenario

The team is the IT department of a small online retailer, the Einzelhändler Company. In addition to supporting the organization's infrastructure, the team is needed to prototype a potential disaster recovery approach using the AWS cloud.

The company has had a topology built in AWS, and now it is time for the IT department to evaluate the management and use of this facility. The IT Director expects the consultants who have built the prototype topology to turn it over to the IT Department on Saturday for the team to begin to exercise and evaluate.

In order ensure the organization is considering the key IT security concerns, they are adopting the Cyber Security Framework, which is an adaptation of elements from:

- NIST 800-53 and its DoD derivative NIST 800-171
- NIST Cybersecurity Framework
- ISO 27001/27002
- CIS Critical Security Controls (CSC)
- Cybersecurity Maturity Model Certification (CMMC)

Midwest Regional Competition Goals

1. To promote fair and equitable standards for cyber defense and technology based competitions that can be recognized by industry
2. To evaluate the defensive and responsive skills of each team under exact hardware, software application, and operating system configurations using a joint academic and industry rating scale
3. To demonstrate the effectiveness of each participating institution's academic security program
4. To be executed by a preponderance of industry professionals
5. To have industry recognition, participation and acceptance of each competition
6. To rate the effectiveness of each competition against a predefined standard of competition rules
7. To provide a cooperative and competitive atmosphere among industry partners and academia in the area of cyber defense education
8. To provide recognition for participating teams

9. To increase public awareness of academic and industry efforts in the area of cyber defense education
10. To select educational teams to represent the Midwest at the National CCDC and National Wildcard CCDC.

Institutional Requirements for Participation

In order to compete at the regional, teams must satisfy the following requirements:

1. Register for the CCDC via the National CCDC registration, or register via Midwest CCDC Registration, and provide roster information to the Midwest CCDC Consortium.
2. All student participants must have submitted a resume in electronic form. These are now collected via National CCDC Registration. Teams that have not registered with the NCCDC must send resumes to the consortium administrator.

Competition Team Identification

- **Blue Team** - student team representing a specific academic institution competing in this competition; Each team consists of up to 12 competitors, declare via National CCDC Registration, or submitted to the competition director as part of their authorization letter. Each competition team may consist of up to eight (8) members chosen from the submitted roster. The minimum team size is four participants. The remainder of the roster is for substitution in the event a member of the active competition team cannot compete. Each competitor is expected to be a full-time student at the school from which they compete. An exception is a part time student that expects to graduate in the current term. Such a student may compete if they were a full-time student during the previous term. \Substitution in the competition team requires approval from the MWCCDC Consortium Competition Director or a MWCCDC Consortium Compliance Monitor present at the competition.

Further guidelines for Blue Team participation have been documented in respective 2025 Midwest State CCDC Team Packets.

  - National rules apply; www.nationalccdc.org
  - Further information is available at https://www.caeepnc.org/mwccdc, the main website for Midwest Collegiate Cyber Defense Competitions.

- **Red Team** – Professional network penetration testers from industry approved by the competition director and industry representatives
  - Scan and map the network of each competition team
  - Attempt to penetrate the defensive capabilities of each Blue Team network and modify any acquired environment
  - Assess the security of each Blue Team network
  - Attempt to capture specific files on targeted devices of each Blue Team network
  - Attempt to leave specific files on targeted devices of each Blue Team network

- Follow rules of engagement for the competition

- **White/Black Team** – Representatives from industry who serve as competition judges (Black Team), remote site judges and remote judges (White Team), scoring management, room monitors and security enforcement in the various competition rooms. Judges will assess the competition team's ability to maintain their network and service availability based upon a business inject and a scoring instrument, delivering inject scenarios, scoring of injects, creating log entries, securing log files, issuing or controlling the timing of injects, etc. Each competing Blue Team may have a White Team member present in their room that will assist judges by observing teams, confirming proper inject completion as well as reported issues.

- **Chief Judge:**
  - Serves as the final authority on scoring decisions or issues relating to equity or fairness of events or activities
  - Cannot be from any institution that has a competing Blue team or have any interest in any team outcome
  - Ideally, should be a representative from industry or law enforcement
  - Final authority of all judging decisions, including assessment of final scores and winners of the competition

- **Gold Team** – Comprised of the MWCCDC Consortium Competition Director, as well as representatives from industry and academia who make up the administration team both in planning and during the exercises. Responsibilities include, but are not limited to,
  - Administration and staffing of the cyber defense competition
  - Works with industry partners to orchestrate the event
  - Along with Industry White Team approves the Chief Judge
  - Has the authority to dismiss any team, team member, or visitor for violation of competition rules, inappropriate or unprofessional conduct

- **Orange Team** – Student workers and professionals who assist the competition by accessing team services. Orange team members act like typical users, and may come from extraneous IP addresses. The Orange Team is expected to keep track of the accessibility of services and report to the White Team their results. The white team will use results of Orange Team activity as a part of scoring.

- **Green Team** – Tech support – assists with any technical needs necessary to maintain the integrity of the competition.

Competition Topology

The competition topology for the 2025 Erich J. Spengler Midwest Regional CCDC comprises twofold network systems as follows:

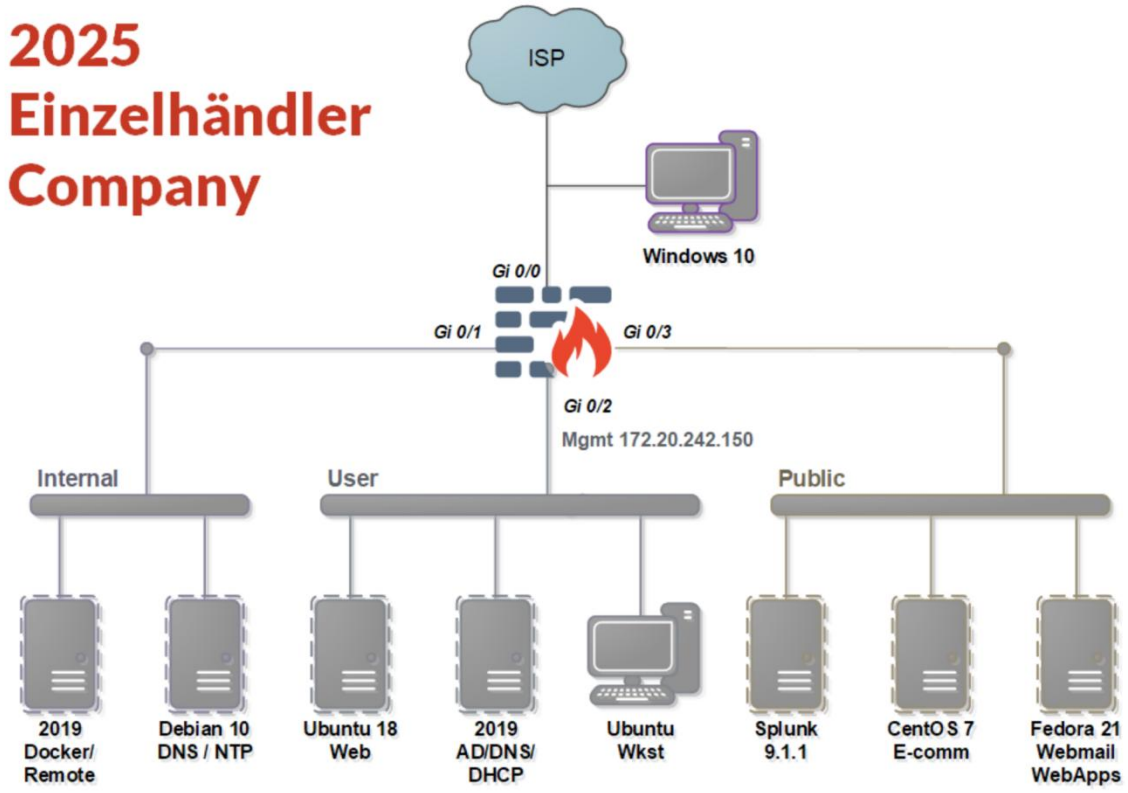
The Einzelhändler Company Virtual Network

The Einzelhändler Company Prototype AWS Cloud

The Virtual Network is all virtual powered by NETLAB. Teams will recognize the Virtual Network as an update to the topology used for the qualification CCDC. The Virtual Network will be used on both Friday and Saturday. There are no public inbound connections to the Virtual Network. Outbound connections may be made via a Web Proxy.

The Prototype AWS Cloud will be brought online Saturday to initiate new services, and provide a possible means to backup critical services provided by the Virtual Network.

The Einzelhändler Company Virtual Topology



- Teams have access to 10 VMs – 7 servers, 2 workstations, and the Cisco Firepower firewall.
- All servers, workstations, and Cisco firewall are virtual machines under the management of NETLAB+™ VE.

- Teams do not have access to the underlying layer 2 switch for the Virtual Network.
- The firewall shown in the topology is a Cisco FTD VM, version 7.2.5.
- You can access the Cisco FTD VM either directly, which yields a command window, or via a browser https://172.20.242.150 from any of the User LAN VMs. The PA user/password are,

  admin/!Changeme123

- Each team has the following Cisco FTD internal addresses:

  Internal, g0/1 172.20.240.254/24
  User, g0/2 172.20.242.254/24
  Public, g0/3 172.20.241.254/24

- Core IP addresses are the following on interface g0/0:

Team	Cisco FTD g0/0 Outbound to Core	Core connection to Cisco FTD	"Public" IP Pool
1	172.31.21.2/29	172.31.21.1	172.25.21.0/24
2	172.31.22.2/29	172.31.22.1	172.25.22.0/24
3	172.31.23.2/29	172.31.23.1	172.25.23.0/24
4	172.31.24.2/29	172.31.24.1	172.25.24.0/24
5	172.31.25.2/29	172.31.25.1	172.25.25.0/24
6	172.31.26.2/29	172.31.26.1	172.25.26.0/24
7	172.31.27.2/29	172.31.27.1	172.25.27.0/24
8	172.31.28.2/29	172.31.28.1	172.25.28.0/24
9	172.31.29.2/29	172.31.29.1	172.25.29.0/24
10	172.31.30.2/29	172.31.30.1	172.25.30.0/24

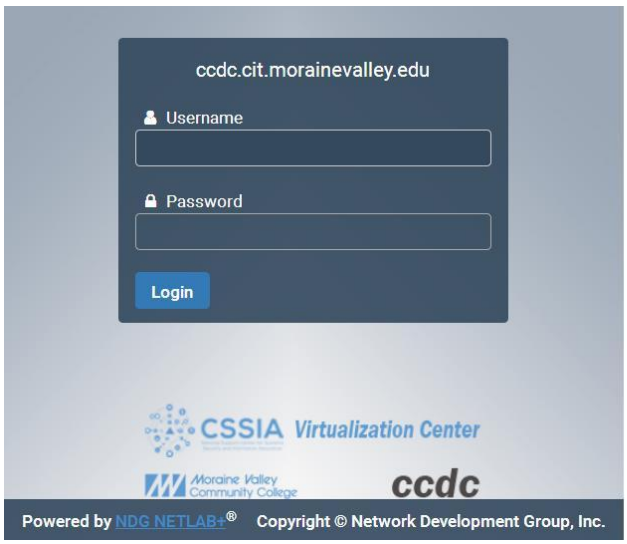
VM data are as follows: This table is accessible on the topology tab of NETLAB+™ VE, via the “Content” upper left.

	Version	*IP*	*Username*	*Password*
INTERNAL				
2019 Docker/Remote	Server 2019 Std	172.20.240.10	administrator	!Changeme123
Debian 10 DNS/NTP	Debian 10	172.20.240.20	root sysadmin	changeme changeme
USER				
Ubuntu 18 Web	Ubuntu Server 18.04	172.20.242.10	sysadmin	changeme
2019 AD/DNS/DHCP	Server 2019 Std	172.20.242.200	administrator	!Password123
Ubuntu Wkst	Ubuntu Desktop 20.04	DHCP	sysadmin	changeme
PUBLIC				
Splunk	9.1.1	172.20.241.20	root sysadmin admin (Web UI)	changemenow changemenow changeme
CentOS 7 E-comm	CentOS 7	172.20.241.30	root sysadmin	changeme changeme
Fedora 21 Webmail/WebApps	Fedora 21	172.20.241.40	root	!Password123
Cisco FTD	FTD 7.2.5	172.20.242.150	admin	!Changeme123
Windows 10	Windows 10	172.31.xx.5	minion	kingbob

Connecting to the Competition Network

- The Virtual Network will be hosted via the Cyber Stadium located at Moraine Valley Community College. This network will be located remotely from any competition room and will be logically isolated from all other competing Blue Teams. Access the NDG NETLAB+™ VE system at,

ccdc.cit.morainevalley.edu



The NDG NETLAB+™ VE is accessible via a web browser.

Accounts to access the Virtual Network will be,

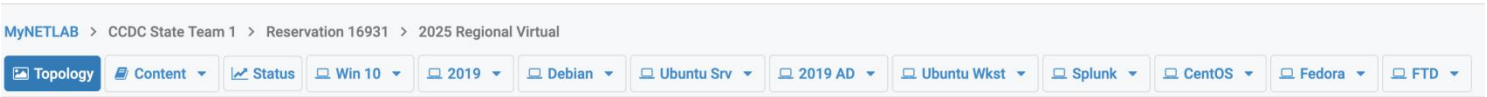
v1u1, v1u2,….,v1u8
v2u1,v2u2,…..
….
v10u1,…

These accounts are of the form v{team#}u1-v{team#}u8.

Each of these accounts allows only one login per account. The password to access the Virtual Network will become valid at the drop flag each day.

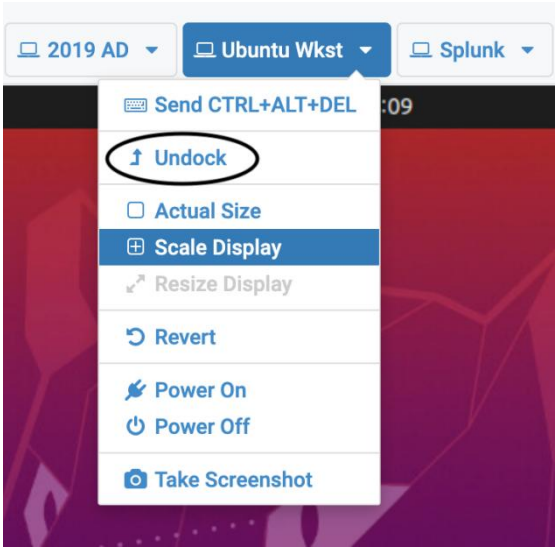
Once logged in on either account, you will need to change your password. **Please remember or document your new password.** Password resets NWTLAB accounts will be via a Tech Support inject.

Teams should see a lab reservation with each respective set of accounts. When entering the topology, the top menu should look similar to the following:



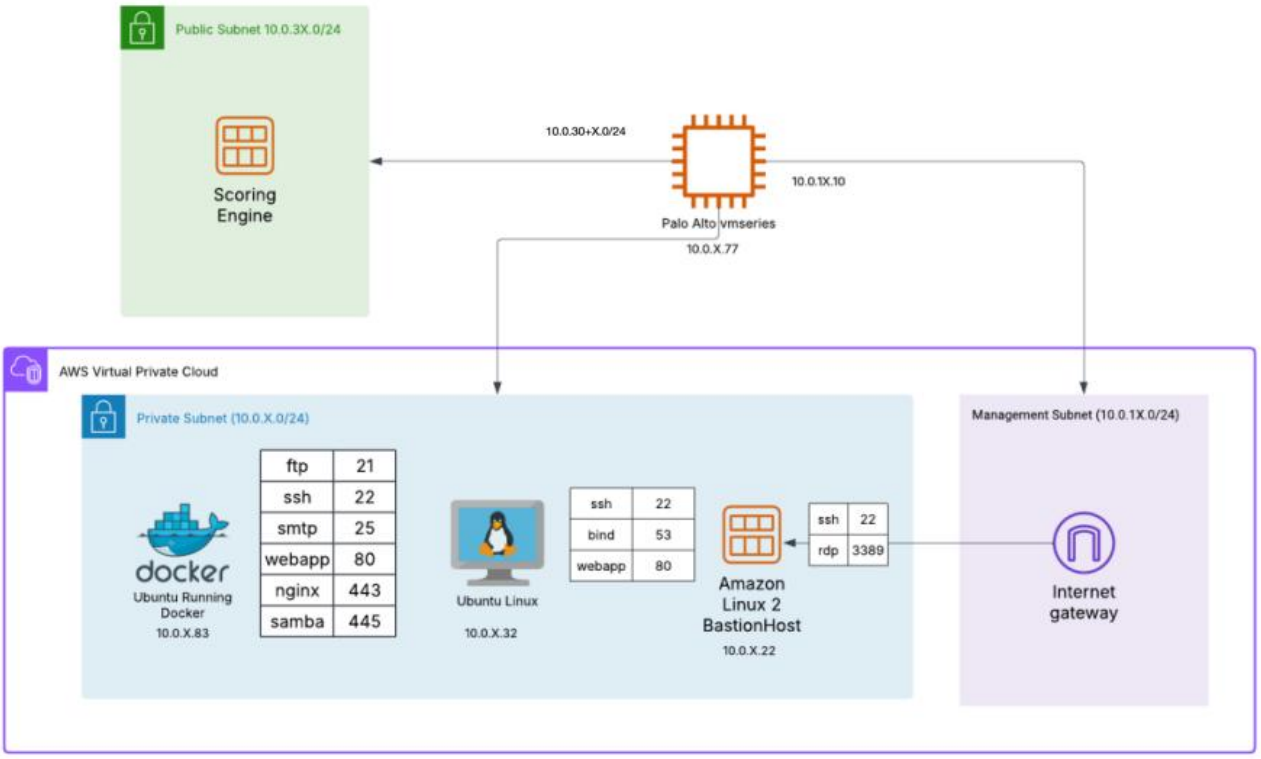
Teams should be cautious with the Reservation drop-down where it's possible to end your reservation. Your system will be restarted with services points retained, but your team will have start once again from the beginning.

Users can still access VMs by clicking on the topology diagram, but they can also click on the appropriate button at the top of the screen shown in the image. The VM access will then replace the topology image instead of a separate window as with a PE system. Users might wish to work on a VM in a separate window which they can do by the 'Undock' feature.



- Teams are provided workstations that adhere to NDG guidelines. See, https://www.netdevgroup.com/products/requirements/#client/
- Each competition network will be physically and logically isolated from the network used to access the system.

Einzelhändler Amazon Web Services Network Topology & Access



Einzelhändler has contracted a migration service provider to move certain services into Amazon Web Services. The migration service provider has created a private subnet for the company in their AWS Virtual Private Cloud. For the time being the migration service provider will manage the underlying AWS fabric, but management of AWS based services is being handed over to the Einzelhändler network group. As such Einzelhändler network personnel will not have access to the AWS management console.

The private subnet contains a Bastion host for console access to the host systems. One host runs a handful of scored support services in Docker Containers and another is provisioned for public Web Services and runs an auxiliary bind DNS server (both will likely be scored).

'Public' traffic is available via the 'Public' subnet in the ranges 10.0.30+X.0/24 where X is the team number. Traffic originating from these networks must be allowed for service checks. Support services from 192.168.100.0/24 must also remain permitted to each host for service delivery needs.

Information on the hosts to be transitioned to the IT teams are as follows:

- Palo Alto VM Series virtual appliance.

Palo Alto VM Series 100 - PAN OS v 10.1.13-h1		
IPv4: 10.0.1X.10	n/a	Management Subnet
IPv4: 10.0.X.77	ethernet 1/1	Private Subnet : zone inside
IPv4: 10.0.30+X.10	ethernet 1/2	Public Subnet : zone outside
Credentials : User - admin Pass - Changeme2!		

The Palo Alto Web GUI interface will be accessible via a public IP disclosed in a business inject. A command line/ ssh session can also be opened via the same public IP.

It's not recommended to update the Palo Alto beyond 10.3.

- Amazon Machine Images:

The delineation of local IP and administrative access credentials per VM are as follows:

Name	Base Operating System	IP address*	Hostname*	Username	password
Ubuntu Web	Ubuntu 24.04.1 LTS	10.0.X.32	ip-10-0-X-22.us-east-2.compute.internal	blueadmin	changeme2
Ubuntu Docker	Ubuntu 16.04.7 LTS	10.0.X.83	ip-10-0-X-83.us-east-2.compute.internal	blueadmin	changeme2
Amazon Linux Bastion	Amazon Linux 2	10.0.X.22	ip-10-0-X-22.us-east-2.compute.internal	blueadmin	changeme2

* X denotes the team number

Ubuntu systems can be accessed once authenticated to the Amazon Linux Bastion Instance via ssh to a public IP which will be revealed in an initial business inject (together with the public IP to your PA) and AWS access will be enabled at a subsequent AWS drop flag. This can be achieved either from direct connection with an underlying ssh session:

```
Ex: [ec2-user@ip-10-0-1-22 ~]$ ssh blueadmin@10.0.1.32
```

Or via establishing local tunnels from the originating host

```
Ex: > ssh blueadmin@3.149.242.87 -L 6000:10.0.1.83:22 -L 5000:10.0.1.32:22 &
      > ssh -p blueadmin@localhost -p6000
      > password:
      blueadmin@ip-10-0-1-83:~$
```


```
Ex: > ssh blueadmin@3.149.242.87 -L 6000:10.0.1.83:22 -L 5000:10.0.1.32:22 &
      > ssh -p blueadmin@localhost -p6000
      > password:
      blueadmin@ip-10-0-1-83:~$
```


Scored Services per host:

Host	IP Address	Service : Port
Ubuntu 24.04.1 LTS	10.0.X.32	JuiceBox : 80
		ftp: 21
		smtp : 25
Ubuntu 16.04.7 LTS	10.0.X.83	web: 80
		bind dns: 53
Amazon Linux 2	10.0.X.22	ssh : 22

Systems are loaded with various user accounts. Teams are responsible to know which accounts are used for services. **Root, Administrator, Admin, Sysadmin, ec2-user, or ubuntu will never be used for scoring.**

The account blueadmin will likely be used for scoring, and since it provides access to the Amazon machine images, it's important to change the password to this account. It's advisable to make the blueadmin account password the same for all three Amazon machines. Look for a separate inject to submit password change requests for the AWS network.

Accessing the Team Portal/ NISE – National Inject Scoring Engine

The Black/White and Red Teams and each respective Blue Team will communicate with each other via a Team Portal, a trouble ticket and response application (NISE – National Inject Scoring Engine) residing within the remote network at Moraine Valley Community College. This system is accessible via a browser with URL,

ccdcadmin3.morainevalley.edu

Note that the MWCCDC Consortium supports an additional NISE/Team Portal,

ccdcadmin1.morainevalley.edu

Follow the instructions from your competition manager for the specific NISE/Team Portal that will be used for your CCDC competition.

Students should login to the NISE first. There are eight accounts per team that may be used to connect to the NISE where multiple logins using the same account is permissible. The accounts for team1 are,

team01a, team01b, team01c, …, team01h

For team2 the accounts are

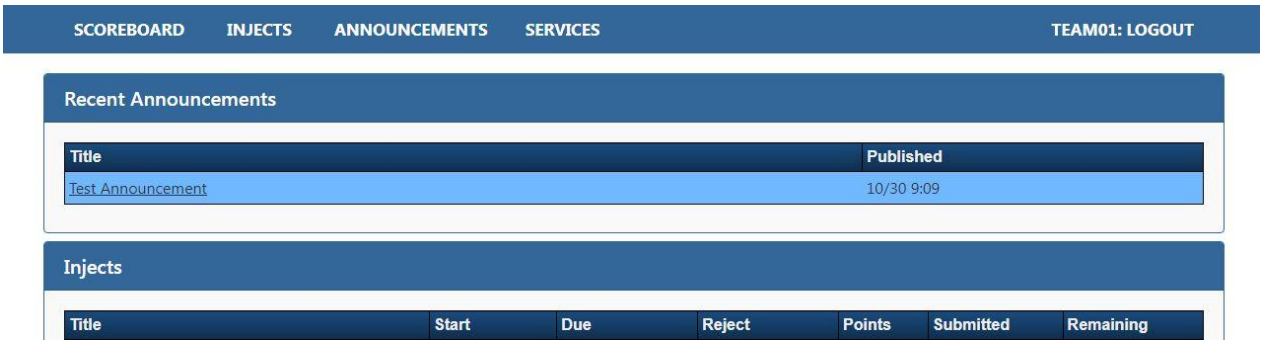
team02a, team02b, …, team02h

It's always two digits for the team number so team10 accounts are,

team10a, ….

Each team account has its own unique password which does not need to be changed.

After logging in the main (Inject) page is displayed.



Using the NISE platform is straightforward and intuitive. Teams should explore the various features of the NISE during the event and be especially attentive to announcements and new inject tasks. **Times displayed on the NISE platform are set to CST.**

Responses to inject tasks must be in the form of an attached PDF file unless directed otherwise. It is possible to attach files with other formats, such as text files, but these cannot be read within the NISE platform. Such files must be downloaded and handled separately.

After an inject submission has been submitted, and while the inject task is still open, both teams and judges may mark a submission invalid allowing the team to resubmit. There is a limit to how many times a team may resubmit, depending on Black/Black/White Team policy.

Teams should be attentive to inject submission deadlines, and the NISE platform may need to be refreshed to keep information current. Teams might also take note a slight delay in the display of services compared with the Scoring Engine.

When first connecting to the NISE, a member of the team should check for an initial inject task, usually identified as “Welcome” or something similar. The task simply requests a response back to the competition judges, signaling that access to the NISE has been successful, and that the responding team is ready to compete.

Once the competition judges have verified that all teams are ready to compete, or have provided ample time to respond, the competition judges will release a notification indicating the drop flag has been issued and the competition has started.

- Teams should be attentive to monitor inject requests and notifications via the Team Portal/NISE.
- Each Blue Team network will be monitored by a scoring system operating within the remote network. An indication of services, as viewed by the indigenous scoring engine, will be made available to each Blue Team via the Team Portal/NISE.
- **SLA will be in effect for all scored services**, meaning that a penalty will accrue if services are down too long.

- **While every effort is made to provide a stable and well defined competition topology, it is subject to change and /or modification as decided by the MWCCDC Consortium Director.**

Functional Services

Certain services are expected to be operational at all times or as specified throughout the competition. In addition to being up and accepting connections, the services must be functional and serve the intended business purpose. At random intervals, certain services will be tested for function and content where appropriate.

HTTP
A request for a specific web page will be made. Once the request is made, the result will be stored in a file and compared to the expected result. The returned page must match the expected content for points to be awarded.

HTTPS
A request for a page over SSL will be made. Again, the request will be made, the result stored in a file, and the result compared to the expected result. The returned page needs to match the expected file for points to be awarded.

SMTP
Email will be sent and received through a valid email account via SMTP. This will simulate an employee in the field using their email. Each successful test of email functionality will be awarded points.

DNS
DNS lookups will be performed against the DNS server. Each successfully served request will be awarded points.

POP3
POP3 connections will be performed against the system using usernames from Active Directory. Once connected a series of commands will be run and the output examined. Correct responses will be awarded points.

FTP
Access to the FTP service either via authentication or anonymous will be made, and the presence of various files will be checked.

TFTP
Access to the tftp service will be made to pull down a small file with an integrity check.

SSH
Access to ssh service with authentication will be made.

Schedule – all times CST

Friday, March 14, 2025

10:00 - 11:30AM - Alumni Hall, SULB	Check-in at Purdue University Northwest, Student Union Library Building (SULB), Alumni Hall (third Floor) Receive Competition Packets, Team and Room Assignment, passwords NISE & competition network access for day 1&2 Team captain signs off team packet Mingle with the Sponsors
11:30AM - Alumni Hall, SULB	David Durkee - Welcome & Introduction to the 2025 Erich J. Spengler Midwest Regional CCDC A Word from our Sponsors
12:30PM - Cafeteria, SULB	LUNCH in the Cafeteria, SULB
1:30PM	Students go to their competition/lab room/ login to NISE/ respond to Welcome inject;
2:30PM	Drop Flag - Competition starts; Scoring is from 2:30-8:30PM
5:30 PM	DINNER No food in the competition/lab rooms!
8:30PM	Scoring ends for the day
9:00PM	All teams must vacate the buildings by this time.

Saturday, March 15, 2025

8:00AM	Student Teams arrive at Purdue University Northwest Go straight to the competition/lab room; AWS Public IP released
9:00AM	Drop Flag for Day 2 - Scoring is from 9:00AM - 6:00PM; AWS drop flag soon follows
12:30PM	LUNCH – Food will be delivered to nearby classrooms, not far from team competition labs No food in the competition/lab rooms!
2:00 - 4:00PM	Team Presentations
6:00PM	Competition ends/Scoring ends DINNER in the Cafeteria SULB
7:00PM	Alumni Hall - Presentations by Red & White Team representative(s) Announce final winners and hand out trophies to 1st, 2nd and 3rd place teams
7:30PM	Competition Director meets with First Place Team

Systems

1. Each team will start the competition with identically configured systems.
2. Teams will be provided access credentials to the Team Portal/ NISE and the Virtual Network by the day of the competition. Public access URL or IP for the AWS will be released on Saturday.
3. Teams should not assume any competition system is properly functioning or secure.
4. Throughout the competition, Green, Orange, and White Team members will occasionally need access to a team's systems for scoring, troubleshooting, etc. Teams should not be alarmed if 'status' shows access to team VMs is being made by those managing the event. Such access will never have a negative impact to teams, and no red team member will have such access.
5. Network traffic generators may be used throughout the competition to generate traffic on each team's network. Traffic generators may generate typical user traffic as well as suspicious or potentially malicious traffic from random source IP addresses throughout the competition.
6. Teams must maintain specific services on the "public" IP addresses assigned to their team. Moving services from one public IP to another is not permitted unless directed to do so by an inject. Likewise, teams are not permitted to change the internal addressing of the competition network unless directed to do so by an inject.
7. Teams are not permitted to alter the system names or IP address of their assigned systems unless directed by an inject; this may affect the results of the scoring mechanism.
8. **Teams are permitted to move services to another platform, provided that the same "public" IP address and DNS naming convention is maintained, along with other requirements of the service.**
9. Teams must maintain "public" services as available from nominally all source IP addresses. Teams may block single IP addresses determined to be the source of malicious traffic. Teams should note that attempts to restrict or filter by IP source address may adversely affect scoring directly. Teams are prohibited from blocking access from entire IP subnets, or from placing a knife edge ACL at the ingress of their networks, attempting to allow only the scoring engine. Teams should be made aware that that controls are put in place and monitored by team based on traffic generation.
10. In the event of system lock or failure, teams will be able to request a restoration (scrub/snapshot) of a particular VM via a Tech Support inject released from the NISE. This applies equally to the Virtual and AWS Networks. **Look for separate Tech Support injects to solicit technical assistance or reset requests on the Virtual and AWS Networks**. A reset will restore any system to its initial starting configuration. The number of system restorations will be tracked and negatively impact scores at the discretion of the Chief Judge. Teams should also consider that system restoration will take time, and that original vulnerabilities reappear.
11. Systems designated as user workstations within the competition network are to be treated as user workstations and may not be re-tasked for any other purpose by teams.

12. Teams may not modify the hardware configurations of workstations used to access the competition network. Teams may not install additional software on the workstations used to access the competition network.
13. Communication with anyone outside the team is prohibited.
14. Teams may not use any external, private electronic staging area or FTP site for patches, software, etc. during the competition. Teams are not allowed to access private Internet-accessible libraries, FTP sites, web sites, network storage, email accounts, or shared drives during the competition. All Internet resources used during the competition must be freely available to all other teams. The use of external collaboration and storage environments such as Google Docs/Drive is prohibited unless the environment was provided by and is administered by competition officials. Accessing private staging areas or email accounts is grounds for disqualification and/or a penalty assigned to the appropriate team.
15. Teams may access and use all Github repositories published by teams and made public. A File Share inject will also be released to help teams share files and images in order to complete documents for inject submission.
16. No peer to peer or distributed file sharing clients or servers are permitted on competition networks unless specifically authorized by the competition officials.

Competition Rules: Acknowledgement & Agreement

Competition rules are applicable to all participants of the Midwest CCDC. They provide structure for the makeup of student teams, permitted actions during competition play, guidelines for scoring, and contingencies for handling disputes. They also document expectations for appropriate conduct during the entire time participants are guests at the host site. Team response to the Welcome inject is tacit acknowledgement of competition rules and their commitment to abide by them.

Team advisers and team captains are responsible for deploying the competition rules to the remaining members of their team. Host sites reserve the right to stipulate additional rules conforming to local policies and guidelines.

Competition Rules: Student Teams

1. Each team will consist of up to no more than eight members. All team advisers have been informed of and will adhere to all national rules. See www.nationalccdc.org
2. Each team may have no more than two graduate students as team members.
3. Team advisers and faculty representatives may not assist or advise the team during the competition. Team advisers and faculty representatives may not be involved in any scoring or decisions that involve a participating institution or Blue Team.
4. Each team will designate a Team Captain for the duration of the competition to act as the team liaison between the competition staff and the teams before and during the competition. **Team Captains should identify themselves to the White Team by team number, and not by institution.**

Competition Rules: Professional Conduct

1. All participants are expected to behave professionally at all times they are visiting the host site, and at all preparation meetings.
2. Host site/ local site policies and rules apply throughout the competition.
3. All Midwest Cyber Defense Competitions are alcohol free events. No drinking is permitted at any time during the competition.
4. Activities such as swearing, consumption of alcohol or illegal drugs, disrespect, unruly behavior, sexual harassment, improper physical contact, becoming argumentative, or willful physical damage have no place at the competition.
5. In the event of unprofessional conduct, student team members and their adviser will meet with Gold Team members upon request. The consequence of unprofessional conduct will be determined by the Site Administrator with the recommendation of the Gold Team. This may be a warning, point penalty, disqualification, or expulsion from the campus.
6. The Site Administrator or a Gold Team member from MWCCDC Consortium reserves the right to disqualify an offender from participation in future competitions.

Competition Rules: Competition Play

1. All requests for items such as software, service checks, system resets, and service requests must be submitted to the White Team. Requests must clearly show the requesting team by number, action or item requested. **Teams should not identify the school they represent to the White Team but communicate by team number.**
2. Teams must compete without outside assistance from non-team members which includes team advisers and sponsors. All private communications (calls, emails, chat, directed emails, forum postings, conversations, requests for assistance, etc.) with non-team members are forbidden and are grounds for disqualification.
3. No PDAs, memory sticks, CD-ROMs, electronic media, or other similar electronic devices, are allowed in the competition unless specifically authorized by the Black/White Team in advance.
4. Printed reference materials (books, magazines, checklists) are permitted in competition areas and teams may bring printed reference materials to the competition.
5. Teams may freely access the list of Github team repository sites during the competition. These will be published during the competition.
6. An unbiased Red Team will probe, scan, and attempt to penetrate or disrupt each team's operations throughout the competition.
7. Teams are permitted to replace applications and services provided they continue to provide the same content, data, and functionality of the original service. For example, one mail service may be replaced with another provided the new service still supports standard SMTP commands, supports the same user set, and preserves any preexisting messages users may have stored in the original service. Failure to preserve preexisting data during a service migration will result in a point penalty as deemed appropriate by the White Team for each user and service affected.

8. Teams are free to examine their own systems but no offensive activity against other teams, the White Team, or the Red Team is permitted. This includes port scans, unauthorized connection attempts, vulnerability scans, etc. Any team performing offensive activity against other teams, the White Team, the Red Team, or any global asset will be immediately disqualified from the competition. If there are any questions or concerns during the competition about whether or not specific actions can be considered offensive in nature contact the White Team before performing those actions.
9. Blue Team members may not change usernames within their respective environment, unless directed to do so by the White Team. Blue Team members may change passwords for administrator and user level accounts. **Changes to passwords affecting scored services must be communicated to the Black/White Team via the password change request feature of the NISE. Look for a password change policy to be issued during the competition. Changes to administrator and root account passwords may be changed without notification, since these are not used for scoring services**. Competitors have the responsibility to determine how accounts relate to services.
10. Teams are allowed to use active response mechanisms such as TCP resets when responding to suspicious/malicious activity. Any active mechanisms that interfere with the functionality of the scoring engine or manual scoring checks are exclusively the responsibility of the teams. Any firewall rule, IDS, IPS, or defensive action that interferes with the functionality of the scoring engine or manual scoring checks are exclusively the responsibility of the teams.
11. Each Blue Team will be provided with the same objectives and tasks.
12. Each Blue Team will be given the same inject scenario at the same time during the course of the competition.
13. Blue Teams may request information from the White Team and Scoring Manager as to why a particular service is not scoring properly. Disclosure of information regarding non-scoring of services is at the discretion of the White Team. Nevertheless, if core system or scoring system faults are discovered, every effort will be made towards corrective action together with modification of scores to maintain equity and fairness.
14. The White Team is responsible for implementing the scenario events, refereeing, team scoring and tabulation.
15. Scoring will be based on keeping required services up, controlling/preventing unauthorized access, and completing business tasks in a timely manner that will be provided throughout the competition.
16. Scores for inject completion and incident reports will be maintained by the White Team, and will not be shared with Blue Team members. Running totals will not be provided during the competition. Some debriefing of a general nature is likely at the end of the competition.
17. If a scenario or event arises that may negatively impact the integrity or fairness of any aspect of the competition that was not previously anticipated, it is the final decision and discretion of the Chief Judge to make adjustments in scores, or deploy new policies.

Competition Rules: Internet Usage

1. Competition systems will have controlled access to the Internet for the purposes of research and downloading patches. **All internet access from the competition network will be through a web proxy.**
2. The web proxy will permit a predetermined set of common web sites including several used for software repositories. Per National CCDC practice, the complete list of accessible sites will not be published.
3. Internet activity will be monitored throughout the competition to assure compliance with competition rules.
4. Additional software or tools must be either free or open source within the limits of predetermined internet accessibility. No software is permitted based on free trial.
5. Teams may not use any external, private electronic staging area or FTP site for patches, software, etc. during the competition other than the published list of Github repositories.
6. All network activity that takes place on the competition network may be logged and is subject to release. **Competition officials are not responsible for the security of any personal information, including login credentials that competitors place on the competition network.**

Competition Rules: Scoring

1. Scoring will be based on keeping required services up, controlling/preventing unauthorized access, mitigating vulnerabilities, and completing business tasks that will be provided throughout the competition. Teams accumulate points by successfully completing injects, maintaining services, and by submitting incident reports. **SLA will be implemented**, meaning teams lose points by violating service level agreements after services have been unavailable, usage of recovery services, and successful penetrations by the Red Team.
2. Scores will be maintained by the White/Black Team. Individual tracking of services for the Virtual Network will be available to respective teams during the competition. Tracking of AWS services may not be available. Blue Team members should use available service reports and internal testing to assess the integrity of their network. Blue Team members should refrain from making direct requests to the White Team for routine service verification.
3. Any team action that interrupts the scoring system is exclusively the fault of that team and will result in a lower score. Should any question arise about specific scripts or how they are functioning, the Team Captain should immediately contact the competition officials to address the issue.
4. Any team that tampers with or interferes with the scoring or operations of another team's systems will be disqualified.
5. Teams are required to provide incident reports for each Red Team incident they detect. Incident reports can generally be completed as needed throughout the competition and submitted to the White Team. The White Team reserves the right to stipulate the times and manner in which incident reports may be submitted. Incident reports must contain a description of what occurred (including source and destination IP addresses, timelines of activity, passwords cracked, etc), a discussion of what was affected, and a remediation plan. The

White Team, in conjunction with the Red Team, will assess scores for incident report submission based on clarity, thoroughness, and accuracy. The White Team may also, at their discretion, assess negative scores for frivolous, unnecessary, or excessive communication.

6. The emphasis on Incident Response reports should be directed towards actual exploitation and unsolicited access. This is in contrast to faults in configuration, bogus accounts, unless caused by exploitation and subsequent action, or even the presence of malware.
7. The winner will be based on the highest score obtained during the competition. Point values are broken down as follows:

35-50%	Functional services uptime as measured by scoring engine
35-50%	Successful completion of inject scenarios will result in varying points, depending upon the importance or complexity of the inject scenario
10-30%	Incident Response and Red Team Exploitation Assessment

Precise percentage breakdown will be determined by the White Team.

Business Tasks

Throughout the competition, each team will be presented with identical business tasks. Points will be awarded based upon successful completion of each business task. Tasks will vary in nature and points will be weighted based upon the difficulty and time sensitivity of the assignment. Tasks may contain multiple parts with point values assigned to each specific part of the tasking. Each business task may have an indication of relative importance or value assigned and a specific time period in which the assignment must be completed. Business tasks may involve modification or addition of services.

Questions and Disputes

1. Team captains are encouraged to work with the MWCCDC Consortium Director, the White/Black Team, and contest staff to resolve any questions or disputes regarding the rules of the competition or scoring methods before the competition begins. Protests by any team will be presented by the Team Captain to the competition officials as soon as possible. Competition Gold Team officials will be the final arbitrators for any protests or questions arising before, during, or after the competition and rulings by the competition officials are final.
2. In the event of an individual disqualification, that team member must leave the competition immediately upon notification of disqualification and must not re-enter the competition area at any time. Disqualified individuals are also ineligible for individual awards or team trophies.
3. In the event of a team disqualification, the entire team will be blocked from the competition environment immediately upon notice of disqualification and is ineligible for any individual or team award.

Aftermath

Members of MWCCDC Consortium, Gold, White, and Green Teams strive to make the Midwest CCDC enriching experiences. All management and administrative teams are open to feedback and suggestions for improvement after the completion of the competition. This may include areas of concern or dissatisfaction.

Whether feedback is positive or negative, participants are forbidden from publishing, posting on the internet, or publicly communicating details of the competition other than what is available at https://www.caeepnc.org/mwccdc. They are also forbidden from publishing, posting on the internet, or publicly communicating assessments of the Midwest CCDC, nor assessments of the performance of any team, nor speculations concerning different possible outcomes. Institutions that fail to adhere to this rule may be refused participation in future competitions.

Institutions may publish, post on the internet, or publicly communicate news stories of a general nature about the Midwest CCDC, and may also enumerate participating teams and winners.

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